MIL-I-44301
21 October 1987

MILITARY SPECIFICATION

INK, MARKING, DRY FOIL, FOR NAME TAPES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of dry foil type ink for application on nylon/polyester tape.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

P-D-245	-	Detergent, Laundry and Hand Dishwashing (Granular)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-T-43664	-	Tape, Textile Nylon/Polyester (for Name Tapes)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7510

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STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard sample. The ink, when applied to camouflage Green 483 polyester tape shall match the standard sample for color and legibility and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

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3.2 Materials. The contractor is given a wide latitude in the selection of materials and manufacturing processes, provided the ink meets all the requirements of this document. The ink shall be a dry foil ink in roll form for use with a hot stamp dry printing system.

3.3 Color. The color of the ink shall be black and when, applied to the specified tape, the impressions shall be dull black equal to color chip No. 37038 of FED-STD-595 when tested as specified in 4.2.3.

3.4 Legibility. The marking ink shall produce a sharp, legible impression on polyester tape when tested as specified in 4.2.3.

3.5 Drying. The ink shall set to touch immediately after printing and shall not smear when rubbed lightly after application when tested as specified in 4.2.3.

3.6 Fastness and legibility of the ink to laundering. The ink, when applied on the specified polyester tape (see 4.3.1), shall have equal or better fastness and legibility than the standard sample after being washed fifty times when tested as specified in 4.2.3.

3.7 Storage life. The ink, after storage for a period of not less than one year, shall be free from grit and shall not deteriorate when tested as specified in 4.2.3.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the dry foil ink shall be furnished in rolls of 600 ± 10 feet on a plastic core, 1 inch ± 0.1 inch inside diameter, 4-1/2 inches wide. No roll shall contain more than one piece. The tape shall be restrained from unwinding by securing the end with a piece of pressure sensitive tape.

3.9 Workmanship. The finished product shall be free from foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item examination. The end items shall be examined for defects listed below. The lot size shall be expressed in units of marking ink rolls. The sample unit shall be one roll of marking ink (600 feet). The inspection level shall be S-2. The presence of any defects on any sample unit shall be cause for rejection of the lot that the sample represents.

Examine	Defect
Ink roll	Not dry foil tape Not color specified Presence of foreign matter
Length	Not as specified, or not in one piece

4.2.3 End item testing. The end items shall be tested for the characteristics listed in table I. The lot size shall be expressed in units of rolls. The sample unit for each test shall be one roll. The lot shall be unacceptable if any roll fails to meet any test requirement specified.

Lot size (rolls)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE I. End item tests

Characteristic	Requirement paragraph	Test method
Color	3.3	4.3.1
Legibility	3.4	4.3.1
Drying	3.5	4.3.2
Fastener and legibility of the ink after laundering	3.6	4.3.3
Storage life	3.7	1/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the requirements stated.

4.2.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content 1/	Number per container is more or less than required.

1/ One shipping container from the sample shall be examined for this defect.

4.2.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.3 Methods of inspection.

4.3.1 Color and legibility. The ink shall be applied to a 1-inch Camouflage Green 483 polyester tape conforming to MIL-T-43664. The printing type shall be Franklin Gothic Condensed 48 point with 1/2 inch letters. Print in one stroke the first ten letters of the alphabet on the tape using a hot stamp printer utilizing a heat cure process. After 1 hour, observe the color and the sharpness of the letters. The dry print shall be compared with color chip No. 37038 for color and dullness.

4.3.2 Drying. Apply the ink as specified above. Immediately after curing touch the printed letters with a finger and examine for transfer of ink. The ink is set to touch if there is no transfer. Also rub letters with a finger and observe smearing.

4.3.3 Fastness and legibility of the ink after laundering. One specimen from the standard sample shall be tested at the same time under the same conditions as the specimen from the sample unit of material being tested.

4.3.3.1 Laundering. One standard size (4-1/2 inch length) printed polyester name tape sample (see 4.3.1) shall be laundered fifty times in accordance with the following procedure.

4.3.3.1.1 Sample preparation. Stitch label onto a piece of broadcloth 2 inches by 6 inches in size. The label and backing shall be folded in half and stitched together so that the print is facing out.

4.3.3.1.2 Washing and drying. Machine wash in a commercial-type washer using warm water ($100^{o} \pm 10^{o}F$) at high water level, on a permanent press cycle using type II detergent conforming to P-D-245 and 4 pounds of ballast. Dry in a commercial-type dryer at low temperature ($100^{o} \pm 10^{o}F$) until dry.

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4.3.3.2 Evaluation. The test specimen shall be compared to the specimen of the standard sample and rated as follows:

Specimen-Pass: Equal or superior to the standard sample. Fail: Inferior to the standard sample. When a failure is reported, the nature of the failure shall also be reported.

5. PACKAGING

5.1 Preservation. Preservation shall be Commercial.

5.1.1 Commercial. Rolls of dry foil tape shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level B or Commercial, as specified (see 6.2).

5.2.1 Level B packing. Ten rolls of dry foil ink, preserved as specified in 5.1, shall be placed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside dimensions of the shipping container shall approximate 10-5/8 inches in length, 10-5/8 inches in width, and 10-1/2 inches in depth. Approximate dimensions are furnished as a guide only. Shipping containers shall be closed and reinforced in accordance with the appendix of PPP-B-636.

5.2.1.1 Weather-resistant fiberboard containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s container fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix.

5.2.2 Commercial packing. Dry foil ink, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), dry foil ink, packed as specified in 5.2, shall be palletized in accordance with load type 1 or 1a of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 104 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable. Shipping containers shall bear the legend:

"STORE IN A COOL DRY PLACE"

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Special markings for type I shelf-life shall be in accordance with the shelf-life markings required in the appendix of MIL-STD-129. Additionally, shipping containers and unitized pallet loads shall be marked as follows:

"Includes Dry Foil Tape, Ten Ea - (NSN)
Expiration Date: "

6. NOTES

6.1 Intended use. The ink is a dry foil type ink to be used for obtaining durable markings on polyester tapes (military name tapes) using a hot stamp printer.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Put up if other than specified (see 3.8).
c. Selection of applicable levels of packing (see 5.2).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.1.1).
e. When palletization is required (see 5.3).

6.3 Standard samples. The standard samples may be obtained from the Commander, U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, Director, Individual Protection Directorate.

6.4 Subject term (key word) listing.

Dry Foil
Ink
Marking

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD
Air Force - 84

Preparing activity:

Army - GL

Project No. 7510-0346

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-44301	Ink, Marking, Dry Foil, For Name Tapes

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP-1, Feb 86

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